Department of Mechanical and Production Engineering
Islamic University of Technology

ME 4879: Robotics

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Course Project - 1

Design, Workspace, and Singularity Analysis of a Robot Manipulator

Overview

In this project you carry a manipulator of your own design from concept through analysis. You will fix its architecture, derive its forward kinematics, map the region of space it can reach, and locate the configurations at which it loses dexterity. The aim is to connect the kinematic theory developed in lecture: degrees of freedom, the homogeneous transformation chain, the Jacobian, and singularities, to a concrete mechanism you can visualize and defend.

1. Choose Your Manipulator

- Select a serial manipulator with between two and six degrees of freedom. A planar 2R or 3R arm admits clean closed-form analysis; a spatial 5–6R arm (optionally with a spherical wrist) is more ambitious and more rewarding. Avoid trivial single-joint mechanisms.
- Fix the joint types (revolute / prismatic), link lengths, and joint limits. Verify the degree-of-freedom count using Grübler's formula, and state the task the arm is intended for (pick-and-place, drawing, welding, etc.).

2. Architecture and Forward Kinematics

- Draw the kinematic diagram and assign frames using the class conventions (x, y, z for the fixed frame; n, o, a for the moving frame). Tabulate the Denavit–Hartenberg parameters.
- Form the link transformations $\mathbf{A}_i$ and the full forward-kinematic chain

$$^{R}\mathbf{T}_{H} \;=\; \mathbf{A}_1\,\mathbf{A}_2\cdots\mathbf{A}_n,$$

  and write the end-effector position and orientation as functions of the joint variables.

3. Workspace

- Determine and plot the *reachable* workspace (every point the tool tip can attain) and, where applicable, the *dexterous* workspace (points reachable with arbitrary orientation).
- A Monte-Carlo sweep of the joint variables—sampling within the joint limits and plotting the resulting tip positions—is an acceptable and recommended method; closed-form boundary curves are welcome for planar arms.
- Comment on how link lengths and joint limits shape the workspace.

4. Jacobian and Singularities

- Derive the manipulator Jacobian $\mathbf{J}(\theta)$ relating joint rates to end-effector velocity, $[\mathbf{D}] = [\mathbf{J}][\mathbf{D}_\theta]$.

- Locate the singular configurations, where $\mathbf{J}$ loses rank ($\det \mathbf{J} = 0$ for a square Jacobian, or rank deficiency otherwise). Classify them (boundary vs. interior; shoulder / elbow / wrist where applicable) and mark them on the workspace plot.
- *Optional:* illustrate the manipulability ellipsoid at a few representative poses, showing how it collapses as a singularity is approached.

5. Tools

You may use any combination of the following; state clearly which tool produced which result:

- SolidWorks or other CAD, for the mechanical model and workspace sweep;
- MATLAB or Python (NumPy / SymPy, Robotics Toolbox, roboticstoolbox-python) for symbolic forward kinematics, the Jacobian, and workspace sampling;
- Gazebo / ROS or CoppeliaSim for simulation and visualization;
- or any equivalent package.

6. Deliverables

- A concise technical report containing: the kinematic diagram and D–H table; the forward-kinematics derivation; the workspace plots; the Jacobian and a complete singularity analysis; and a short discussion relating the results to the arm's intended task.
- Supporting artifacts: your code and/or CAD files, together with a brief animation or set of rendered frames showing the arm moving through its workspace and at a singular pose.
- The project may be undertaken individually or in small groups, as arranged with the instructor.

Notation. Use the frame and transformation conventions of Lectures 1–4: $\mathrm{Rot}(\mathrm{axis}, \theta)$ and $\mathrm{Trans}(d_x, d_y, d_z)$; fixed axes x, y, z and moving axes n, o, a; pre-multiplication for current-frame motions and post-multiplication for fixed-frame motions; and superscript-prefix frame notation, e.g. ${}^{U}\mathbf{T}_{R}$.